

# Phaneendra Gavara

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## SUMMARY

Software/AI Engineer with production experience across agentic AI systems, machine learning, data engineering, and full-stack development — including a solo-architected, AWS-deployed agent-native platform, an automated LLM/RAG evaluation pipeline, and multiple production data and web applications. M.S. Data Science graduate of Arizona State University (GPA 3.90/4.0), with current research experience applying AI to blockchain security, scalability, and efficiency.

## EDUCATION

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| <b>Arizona State University</b><br><i>M.S. Data Science (Computing &amp; Decision Analytics Concentration) — GPA: 3.90/4.0</i><br>• <i>Relevant Coursework:</i> Statistical Machine Learning, Data Processing at Scale, Data Mining, Computing & Data-Driven Optimization, Analyzing Big Data, Data Science Capstone | Tempe, AZ<br>Aug 2024 – May 2026          |
| <b>IIIT Bhubaneswar</b><br><i>B.Tech in Computer Science and Engineering — GPA: 7.14/10</i>                                                                                                                                                                                                                          | Bhubaneswar, India<br>Dec 2020 – May 2024 |

## TECHNICAL SKILLS

**Agentic AI & LLMs:** LangGraph, LangChain, Claude API, GPT-4o, Model Context Protocol (MCP), Tool/Function Calling, Multi-Agent Orchestration, RAG, Prompt Engineering, Memory & Context Management  
**ML, Deep Learning & Evaluation:** PyTorch, scikit-learn, CNNs, DistilBERT, SARIMAX, OR-Tools, SHAP, RAGAS, Automated Evaluation Suites, Quality Gates  
**Data Engineering & Analytics:** ETL Pipelines, Star Schema Design, Azure SQL, Azure Blob Storage, PostgreSQL, Pinecone, Pandas, NumPy, Matplotlib, Seaborn  
**Full-Stack Development:** FastAPI, Pydantic, Node.js, React 18, Next.js 14, REST API Design, JWT Auth, RBAC  
**Cloud, DevOps & IaC:** AWS (EKS, Lambda, S3, EC2), Azure (SQL, Blob, AI Search, Key Vault), GCP BigQuery, Docker, Kubernetes, Terraform, Bicep IaC, GitHub Actions/CI-CD, Vercel  
**Languages:** Python, SQL, TypeScript, JavaScript, C++, R, Go

## PROJECTS

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| <b>AgentForge — Agent-Native Platform</b><br><i>Python, FastAPI, React, TypeScript, AWS (EKS, Lambda, S3), Docker, PostgreSQL</i> <ul style="list-style-type: none"><li>Solo-architected and built a production agent-native platform (“Vercel for AI agents”) to build, deploy, evaluate, and govern autonomous AI agents, spanning 5 microservices and ~16,800 lines of code</li><li>Deployed the full platform live on AWS EKS behind HTTPS and a WAF, then automated a complete teardown to \$0 idle cost, demonstrating end-to-end cloud lifecycle ownership</li></ul> | 2026 |
| <b>ASU LLM Evaluation Pipeline</b><br><i>Python, LangChain, RAG, RAGAS, GPT-4o, Pinecone, GitHub Actions, React</i> <ul style="list-style-type: none"><li>Built an automated CI/CD evaluation pipeline that scores a university RAG chatbot across 100 test questions on every code push, enforcing 6 automated quality gates before deploy</li><li>Cut LLM-judge evaluation cost by 25x and sped up result aggregation by 8x through batched scoring and caching</li></ul>                                                                                                 | 2026 |
| <b>InsightHub — AI-Powered Business Analytics Platform</b><br><i>Python, FastAPI, Azure SQL, Azure Blob Storage, Azure OpenAI (GPT-4o), Azure AI Search, React</i> <ul style="list-style-type: none"><li>Designed and deployed a full-stack Azure analytics platform ingesting 230,000+ rows through a star-schema ETL pipeline into Azure SQL</li><li>Built a FastAPI backend serving business insights to a React dashboard, fully deployed and live in production</li></ul>                                                                                              | 2026 |
| <b>Republic of Bean — Agentic AI Ordering System</b><br><i>LangGraph, Claude API, FastAPI, PostgreSQL, Model Context Protocol (MCP)</i> <ul style="list-style-type: none"><li>Built a multi-agent conversational ordering system using LangGraph with Claude as the orchestrating agent across specialized sub-agents</li><li>Exposed PostgreSQL-backed internal ordering APIs to the LLM through the Model Context Protocol (MCP), decoupling agent reasoning from application logic</li></ul>                                                                             | 2025 |
| <b>TravellIQ — AI Travel Itinerary Generator</b><br><i>Python, Flask, OR-Tools, scikit-learn, SARIMAX, GPT-4o</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026 |

- Built an AI travel planner that generates a complete, feasibility-checked itinerary in under 90 seconds, combining ML recommendations, image generation, and route optimization
- Modeled demand and pricing trends with SARIMAX time-series forecasting and scikit-learn to inform itinerary recommendations

### **Industrial AI Copilot for Equipment Monitoring**

2026

*Next.js, React, TypeScript, Python, SWR*

- Built a production real-time anomaly detection system for industrial equipment with sub-second alerting
- Built a full automated incident response pipeline to route alerts without manual triage

### **Anomaly Detection in Crowds**

2025

*Python, PyTorch, CNNs, OpenCV, Deep Learning*

- Built a CNN-based crowd safety system in PyTorch that learns normal crowd behavior from video and flags anomalies in real time
- Generated frame-level heatmaps to visualize and localize anomalous activity for rapid human review

### **AI-Powered Developer Portfolio**

2026

*Next.js 14, TypeScript, Tailwind CSS, Framer Motion, GPT-4o mini, Pinecone*

- Built a Next.js 14 developer portfolio with 4 live-switching themes and Framer Motion animations
- Built “Sparky,” an embedded AI assistant powered by GPT-4o mini and a Pinecone RAG pipeline that answers visitor questions about the portfolio’s projects

### **Heart Disease Detection**

2024

*Python, scikit-learn, Pandas, Matplotlib, Seaborn, SHAP*

- Built a supervised ML classifier achieving 85% accuracy on the Cleveland Heart Disease dataset
- Benchmarked multiple model architectures and used SHAP feature analysis to explain individual predictions and surface the strongest risk factors

### **Shaded Route Planning**

2024

*Python, Image Segmentation, Solar Geometry, NumPy*

- Built a city-scale shade simulation that scores pedestrian walking routes by sun coverage across different times of day
- Combined image segmentation and solar-geometry modeling to estimate shade coverage without requiring live sensor data

### **AI Voice Turing Test Model**

2024

*Python, Transformers, NLP, Audio Processing, Librosa*

- Built an audio ML + NLP classifier that distinguishes human from AI-generated calls while analyzing classification behavior for demographic bias, placing 3rd at the ASU Social Bias Hackathon
- Extracted acoustic features with Librosa and fine-tuned a Transformer-based classifier for real-time inference

## **EXPERIENCE**

### **Visiting Researcher**

Aug 2026 – Present

*Arizona State University*

*Tempe, AZ*

- Designing, developing, and implementing AI-based approaches and solutions for blockchain technologies
- Enhancing and analyzing machine learning models, algorithms, and intelligent blockchain applications
- Researching and implementing AI techniques to improve blockchain security, scalability, and efficiency

### **Teaching Assistant & Grader**

Aug 2025 – May 2026

*Arizona State University, Ira A. Fulton Schools of Engineering (CSE 230 – Computer Architecture)*

*Tempe, AZ*

- Graded student submissions and gave feedback across cache memory calculations, AMAT, MIPS multi-cycle control signals, and pipeline hazards using structured, leniency-calibrated rubrics
- Held office hours and managed course communications (Canvas announcements, student Q&A) for a Computer Architecture course

## **CERTIFICATIONS**

AWS Certified Solutions Architect – Associate (Jul 2026)

AWS Certified AI Practitioner (Jun 2026)

AI Fluency Framework & Foundations – Anthropic (May 2026)

Claude Code 101 – Anthropic (May 2026)

Claude 101 – Anthropic (May 2026)

Introduction to Claude Cowork – Anthropic (May 2026)

## **LEADERSHIP & AWARDS**

### **Admin Assistant & Executive Member**

Jan 2021 – Present

*Help N Hope Charitable Society (NGO)*

*Andhra Pradesh, India*

- Co-founded the organization and was appointed Executive Member (2024) for a sustained commitment to making quality education accessible to underprivileged children
- 3rd Place, Social Bias Hackathon – Ethical Spectacle Research x Arizona State University (Oct 2024)